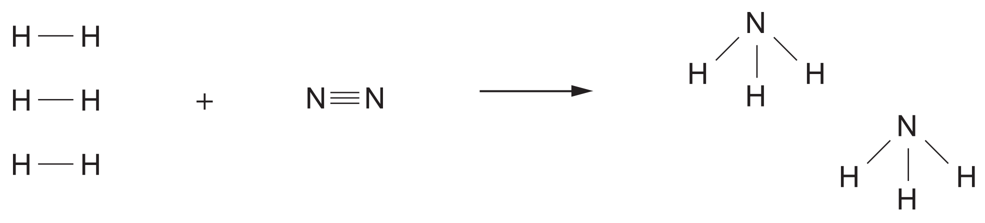
Examiner only

8. Ammonia is manufactured from hydrogen and nitrogen in the Haber process.

(a) The equation shows the bonds which are broken and the bonds which are formed in the production of ammonia.



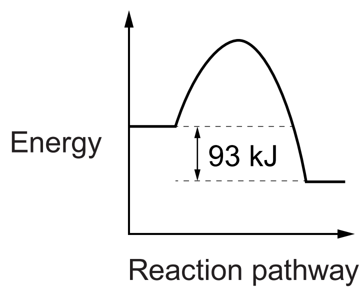
hydrogen nitrogen ammonia

The bond energy of a H—H bond is 436 kJ. The energy needed to break **all** the bonds in the **reactants** is 2253 kJ.

(i) Calculate the energy needed to break a N≡N bond. [2]

Energy = ... kJ

(ii) The diagram below is the energy profile for the reaction.



Calculate the energy released when **all** the bonds in the ammonia molecules are formed. [1]

Energy = ... kJ

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(b) The table below shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)			
	200	300	400	500
	Percentage yield of ammonia			
100	81.7	52.5	25.2	10.6
200	89.0	66.7		18.3
400	94.6	79.7	55.4	31.9
1000	98.3	92.6	79.8	57.5

(i) Describe the relationship between the percentage yield of ammonia and the temperature. [1]

....

(ii) **Circle** the most likely percentage yield of ammonia if the process were carried out at 200 atm and 400 °C. [1]

20 % **30 %** **40 %** **50 %**

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(c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

Compound	Flame test	Add $NaOH(aq)$	Add $HCl(aq)$	Add $BaCl_2(aq)$	Add $AgNO_3(aq)$
A	no colour	pungent gas given off turns damp red litmus paper blue	no reaction	no reaction	white precipitate forms
B	green flame	blue precipitate forms	no reaction	white precipitate forms	no reaction
C	lilac flame	no reaction	gas given off turns limewater milky	no reaction	no reaction

Use the information provided to identify compounds **A**, **B** and **C**. [3]

A ....

B ....

C ....

8

END OF PAPER